The Silicon Sample Tournament: A multi-team benchmark of AI approaches for predicting behavioral-experiment effects

2026-06-11

This is an amendment to our preregistration *"Predicting the effects of behavioral interventions on humans with LLMs"*. Writing the initial preregistration, we conceived of the project as a single-team simulation project. With this amendment, we turn the project into a **multi-team prediction tournament**: we invite independent research teams to predict the results of a novel behavioral megastudy (16 message interventions to increase trust in climate scientists, 13 preregistered outcomes, a representative U.S. sample of ~18,000) using any AI-based approach they choose—behavioral clones, group-level simulation, or direct effect forecasting. Submissions enter at one of **three tiers** defined by the granularity of the prediction (individual-level synthetic data, cell-level statistics, or effect-level estimates with prediction intervals), and every approach is scored on the preregistered metric ladder against the same human–human ceiling. The initial preregistration remains valid: it specifies in detail how we measure the (dis)similarity between the human sample and simulated samples, and all analytic procedures laid out there are kept in place. Our own simulations, registered with the initial preregistration, are reported as one team's approach under identical rules. At the time of writing this amendment, human data are partially collected (~10,000 of the ~18,000 responses); none of these data have been shared, published, or made accessible to any prospective team, and no team will be granted access before the prediction lock. Before any human data are released, every team registers its approach using a standardized method-registration form (an extended version of the GUIDE-LLM reporting checklist) and deposits its predictions on Zenodo—publicly, or partly escrowed under a disclosure policy that keeps every entry locked and verifiable even where it cannot be fully public.

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```r
# packages
library(tidyverse)
library(sandwich)
library(lmtest)
library(marginaleffects)
library(broom)
library(MetBrewer)
library(patchwork)
library(gt)
library(here)

# custom functions (reused unchanged from the original preregistration)
source(here("R/functions/statistics.R"))
source(here("R/functions/plots.R"))
source(here("R/functions/tables.R"))
source(here("R/functions/reporting.R"))

# simulated (random) placeholder data
human_data     <- readRDS(here("data/simulation_preregistration/human_data_preregistration.ro
llm_data_teams <- readRDS(here("data/simulation_preregistration/llm_data_preregistration_tean
```

```r
# Same preregistered random split of the human sample as in the original
# preregistration. Human 1 is the reference half all submissions are scored
# against; Human 2 is the human-human ceiling.
set.seed(42)
split_ids    <- sample(unique(human_data$id), size = floor(length(unique(human_data$id)) / 2
human_data_1 <- human_data |> filter(id %in% split_ids)
human_data_2 <- human_data |> filter(!id %in% split_ids)
```

> ⚠ Warning
>
> **This is an ongoing draft, to share with collaborators. The amendment is not yet registered and will change still.**

> ❗ Important
>
> **This is an amendment.** It builds on, and does not replace, the original preregistration (Preregistration). All analytical procedures are kept in place. This document adds (i) the tournament design with three submission tiers, (ii) a standardized method-registration framework and disclosure policy, (iii) per-tier submission specifications, and (iv) a new cross-team comparison.
> All datasets used in this amendment are simulated random noise with the correct structure.

Motivation

Whether large language models can stand in for human participants is one of the most consequential open questions in behavioral science. The evidence so far is mixed and hard to interpret: most existing evaluations target experiments that were already published—and plausibly in the models' training data—and almost none are preregistered, so reported successes are entangled with researcher degrees of freedom in model and prompt choice (Cummins 2025).

The original preregistration evaluates a single approach to simulating human survey responses with LLMs—our own behavioral clones, built from demographically representative U.S. profiles and a particular model and prompt. However, a single team's result only represents one point in a large design space. A more systematic way of assessing how well AI approaches can predict human behavior in survey experiments is to pool expertise and compare different approaches on the same target.

We will therefore launch a call for a **multi-team prediction tournament**—the *Silicon Sample Tournament*. Predicting a large survey experiment (N 18,000) designed to increase trust in climate scientists provides a unique benchmarking opportunity. The experiment tests the effects of 16 text-based messages[1], and includes both self-reported attitudes (e.g., support for funding of climate science) and behavioral measures (e.g., donations to a scientific organization, a newsletter sign-up) as outcomes. Importantly, none of the interventions has ever been tested in this form. This matters, because the perhaps most discussed use-case of simulated participants is predicting novel research paradigms that would otherwise be very costly to explore with humans. The tournament removes the two problems of existing evaluations by construction: the target is genuinely novel, and every prediction is deposited and time-stamped before any human data are released. What a team predicts is what a team is scored on.

The first goal of the tournament is *not* to crown a single champion team. Per the original preregistration, we evaluate approaches on a ladder of metrics of increasing strictness—from directional agreement to formal equivalence—plus distributional fidelity and stereotyping diagnostics. We expect different approaches to win on different rungs; the point is to map the design space, not to declare one winner. Second, and perhaps more importantly, we aim to provide a **catalogue**: a structured, comparable description of how research teams currently approach AI prediction of behavioral experiments—open- vs. closed-weight models, persona sources, administration protocols, fine-tuning, in-context learning—and which of these choices appear to matter.

The human target sample

Predictions are scored against the human data from the megastudy itself. Participants are recruited from a national, non-probability opt-in panel of U.S. residents (CloudResearch), all aged 18 or older, who pass a series of attention and bot-detection checks before treatment assignment, so that the analyzed sample is relatively attentive to the intervention materials.

The benchmark targets **N 18,000 complete responses**: 1,000 per intervention across the 16 text-based interventions, plus 2,000 in the shared control condition. Recruitment uses **census-based cross quotas** on gender × age and gender × race/ethnicity to approximate the U.S. adult population; the targets below match the proportions of the parent megastudy, with counts simply rescaled to N 18,000. Participants selecting "Other" as their gender are not quota-constrained.

The targets below are drawn from the 2024 U.S. Census Bureau Population Estimates Program. *Total* is the share within each category; *Male* and *Female* give the within-category gender split.

```
read_csv(here("data/quotas_18000.csv")) |>
  gt(groupname_col = "Variable", rowname_col = "Category")
```

[1] The original experiment contains 20 interventions, but we will only use 16, as the other four are interactive interventions that would make the simulation process more complicated.

Table 1: Sampling quota targets (N 18,000), from the 2024 U.S. Census Bureau Population Estimates Program. Proportions match the parent megastudy, with counts rescaled to N 18,000.

	Total	Male	Female
Age			
18-29	3629 (20.2%)	1848 (50.9%)	1781 (49.1%)
30-44	4688 (26.0%)	2365 (50.5%)	2323 (49.5%)
45-59	4122 (22.9%)	2048 (49.7%)	2074 (50.3%)
60+	5561 (30.9%)	2566 (46.1%)	2995 (53.9%)
Race / Ethnicity			
Asian / Asian American	1201 (6.7%)	568 (47.3%)	633 (52.7%)
Black / African American	2212 (12.3%)	1042 (47.1%)	1170 (52.9%)
Hispanic / Latino	3263 (18.1%)	1646 (50.5%)	1617 (49.5%)
Other	492 (2.7%)	240 (48.8%)	252 (51.2%)
White (non-Hispanic)	10832 (60.2%)	5332 (49.2%)	5500 (50.8%)

Gender, age, and race are the only mandatory survey items (needed to fill the quotas) and so enter every analysis model as covariates with no missing data. The six demographic moderators used for the subgroup and calibration analyses—gender, age, race, education, income, and partisan identity—are documented in the original preregistration and codebook; full sampling and exclusion detail is in the human study preregistration.

Tournament design

How it works

Participation is by invitation to teams with a track record in simulating human participants or forecasting experimental results. The procedure for each team is:

1. **Accept the invitation.** Confirm the team and receive the materials package: the intervention texts, the full survey instrument, the codebook, the registration template, and the submission validator. No participant pool is shipped—each team constructs its own profiles (see C.4).
2. **Build the pipeline.** Use any AI-based approach: any models, personas, prompts, fine-tuning, retrieval, or prior literature. Teams never receive any human outcome data.
3. **Register & lock.** Complete the method-registration form (see *Method registration*), generate the predictions, and deposit both on Zenodo—publicly, or partly escrowed under

the disclosure policy. The deposit is hashed and time-stamped before the prediction lock date.

4. **Scoring & publication.** Once human data collection completes, every submission runs through the preregistered analysis pipeline unchanged. Results are reported on a public leaderboard alongside the human–human ceiling, and in a joint manuscript with all participating teams as co-authors.

Ground rules

- **Blinding is absolute.** The human outcome data are collected and held exclusively by the core team. At the time of registering this amendment, only about 10,000 of the planned ~18,000 human responses have been collected, and none of the outcome data have been shared, published, or otherwise made available. No team may access, solicit, or be shown any human outcome data from this study—including pilots—before the prediction lock. A signed attestation is part of registration (item G.3).
- **AI-based and automated.** Pipelines must be AI-based and run without humans in the loop at prediction time. Human effort goes into designing the pipeline, not into adjusting its outputs.
- **Otherwise unrestricted.** Any models (open or closed weight), any prompting or persona strategy, any external training data, prior studies, or meta-analytic knowledge may be used—as long as it is declared (Sections F/G of the registration form).
- **One primary submission per team.** Optional secondary submissions (ablations, alternate models) are welcome and analyzed as clearly labeled exploratory entries.
- **Everything is registered—not everything must be public.** The method form, the prediction files, and—strongly encouraged—code and raw model output logs are deposited on Zenodo (conveniently via a release on a linked GitHub repository) before the lock date. Teams with proprietary methods may seal parts of their registration under the disclosure policy: locked and verifiable is mandatory, public is not.

Three ways to compete

We deliberately welcome very different approaches. Submissions enter at one of **three tiers**, defined by the granularity of the prediction. The common currency across tiers is the set of preregistered estimands—the same quantities the human analysis produces—so every approach is scored on the rungs it can reach. Teams choose the highest tier their approach supports; lower tiers are simply eligible for fewer analyses.

- **Tier 1 · Full — individual-level synthetic data.** A respondent-per-row dataset of behavioral clones: each synthetic participant completes the experiment in one condition. Flows through the preregistered analysis pipeline completely unchanged. Eligible for everything: ATE recovery, calibration regression, response distributions, subgroup heterogeneity, demographic baselines, and stereotyping diagnostics. *Grain: 1 row = 1 synthetic respondent.*

- **Tier 2 · Cell-level — group-level predictions.** Predicted cell statistics—mean, SD, and effective N—per condition × outcome, and per condition × moderator-level × outcome. For approaches that simulate or reason about groups rather than individuals. Eligible for ATEs, calibration, subgroup heterogeneity, and demographic baselines; not eligible for distribution-shape metrics (OVL, KS) or predictability regressions. *Grain: 1 row = 1 cell (condition × [moderator level ×] outcome).*
- **Tier 3 · Effect-level — direct effect predictions.** One predicted ATE per intervention × outcome, in original outcome units, with a **95% prediction interval** expressing the team's epistemic uncertainty. For zero-shot forecasting, in-context learning over the literature, meta-analytic priors, or hybrid pipelines. Eligible for directional agreement, Spearman , Pearson r, RMSE, and calibration regression; inferential agreement and TOST are computed from the submitted intervals. *Grain: 1 row = 1 intervention × outcome estimate.*

Tip

Dual entries encouraged. A team may submit a Tier 1 dataset as its primary entry and a Tier 3 direct prediction from the same pipeline as a secondary entry. Comparing a team's simulated-then-derived effects against its own direct forecasts isolates the value of the simulation step itself—one of the most interesting questions this tournament can answer.

Analysis eligibility by tier

Preregistered analysis	Tier 1	Tier 2	Tier 3
ATE recovery — six-metric ladder (Section 1)			*
Calibration regression / (Section 1)			
Response distributions — variance ratio, OVL, KS (Section 1)		—	—

Preregistered analysis	Tier 1	Tier 2	Tier 3
Subgroup heterogeneity — condition × moderator (Section 2)			—
Within-subgroup distributions (Section 2)		—	—
Demographic baseline calibration (Section 3)			—
Demographic predictability / stereotyping (Section 3)		—	—

* For Tier 3, inferential agreement and equivalence (TOST) are evaluated against the submitted 95% prediction intervals rather than model-based standard errors. For Tiers 1–2, standard errors fall out of the fitted models (or are computed from the submitted cell SDs and effective Ns), so no separate uncertainty input is required.

Method registration

Every team registers its approach using a structured form before the prediction lock. The form builds on the GUIDE-LLM reporting checklist (Feuerriegel et al. 2026)—the consensus reporting standard for LLM use in behavioral science—and extends it where a reporting checklist alone is not sufficient for, in principle, full replicability of a prediction pipeline. Tournament-specific extensions are marked ; item codes A–G follow GUIDE-LLM.

Items concerning human participants or sensitive-data privacy (GUIDE-LLM D.1) are generally *not applicable* here, since every submission is fully synthetic; teams mark these N/A. When multiple models serve different pipeline stages, the model sections are completed separately per model.

> Note
>
> **Replicability standard.** An independent team with access to a team's registration—and nothing else—should in principle be able to re-run the pipeline and obtain a statistically equivalent submission. Where exact replication is impossible (deprecated APIs,

hardware nondeterminism), the deposited raw output logs (item I.2) take over as the verifiable record. Where disclosure is restricted (see *Disclosure policy*), this standard applies to the escrowed registration rather than the public one.

The full item-by-item form—items 0.1 through I.3, with the tournament-specific extensions marked —is published in the call for collaboration. A fillable template (Markdown + machine-readable YAML) ships in the tournament repository.

Disclosure policy — proprietary approaches

We expect some participants—including companies—to be unable or unwilling to publicly disclose their full approach. The tournament accommodates this by separating two things that are usually conflated: **verifiability** (is the registration locked, time-stamped, and auditable?) and **publicity** (can everyone read it?). Verifiability is non-negotiable; publicity is graded. Each registration item is assigned one of three statuses:

- **Public** — included in the open Zenodo deposit, visible to everyone.
- **Escrowed** — deposited in a restricted-access Zenodo record (or equivalent sealed deposit). The record still carries a DOI, a timestamp, and public SHA-256 hashes of its contents, so the lock is publicly verifiable; the contents are accessible only to the core team and up to two designated independent auditors, under a confidentiality agreement, solely for verifying compliance with the tournament rules. An embargo with a sunset date (e.g., public release upon publication, or after 24 months) is strongly encouraged but not required.
- **Withheld** — not deposited at all, not even under escrow. Permitted only for items outside the non-waivable set below, and only at the cost of entering as a sealed entry (Class C).

The overall disclosure class of an entry follows from its most restricted item:

Class	Definition	Consequences
Class A · Open	All registration items public.	Full leaderboard standing; entry is independently replicable in principle; all features enter the approach catalogue and the design-choice analysis.
Class B · Escrowed	Some items sealed, but every item is available to the core team and auditors under confidentiality.	Full leaderboard standing with an *escrowed* badge; compliance is verified but independent replication awaits the embargo sunset; only publicly disclosed features enter the design-choice analysis (the rest coded as *undisclosed*).

Class	Definition	Consequences
Class C · Sealed	One or more permitted items withheld even from escrow.	Scored and reported on the leaderboard with a *sealed / not independently verifiable* flag; excluded from the approach catalogue and the design-choice analysis; the manuscript reports the entry as a performance claim that the core team could not audit beyond the non-waivable core.

The non-waivable core. The following can never be escrow-only-skipped or withheld, because without them scores are uninterpretable for any reader—they protect the validity of the benchmark itself, not the team's IP:

- **Always public:** team identity and contact (0.1), a plain-language summary at a functional level (0.2 — what the pipeline does, not how), tier and approach family (0.3), coverage (0.5), automation confirmation (A.2), competing interests (G.1), the blinding attestation (G.3), the metadata file, and the prediction files themselves.
- **At minimum escrowed** (may be sealed from the public, never from the core team): the external-human-data declaration (G.2), the contamination note (G.4), the internal selection procedure (H.1), and the raw output logs (I.2). These are precisely the items a team gaming the benchmark would want to hide—which is why even Class C entries must place them in escrow.
- **All remaining items** (model details B, prompts and administration C, stochasticity D, post-processing E, learning corpora F, code I.1, resources I.3) may be escrowed freely, or withheld at the cost of Class C.

For commercial teams, the value of participating runs the other way: this is an adversarially designed, preregistered, third-party benchmark on genuinely unseen human data—a substantially stronger credibility signal than any internal evaluation, available at whichever disclosure class the team's IP constraints allow. The leaderboard reports performance regardless of class; the badge tells readers exactly how much of the claim was auditable.

Submission specification

A submission is a Zenodo deposit (ideally generated from a release on a linked GitHub repository, which offers a convenient option to create time-stamped deposits) containing: the completed registration form, the prediction file(s) for the team's tier, a machine-readable metadata file, and—per items I.1–I.2—code and raw output logs. Under the disclosure policy the deposit may be split into a public record and a restricted-access (escrowed) record; the prediction files and metadata are always in the public record, and the public record lists the DOI and content hashes of the escrowed one. Everything must be deposited and time-stamped before the prediction lock date; the DOIs and SHA-256 hashes of all files are emailed to the core team by the same deadline.

Common requirements (all tiers)

- **File naming:** `<team_id>_T<tier>_<primary|secondary-k>_v<n>.csv` — e.g., `vienna_T1_primary_v1.csv`. UTF-8, comma-separated, period decimal separator, `NA` for missing.
- **Coverage:** all 16 simulated interventions plus control, all 13 outcomes. Declared subsets are permitted (item 0.5); uncovered cells are excluded pairwise and coverage is reported on the leaderboard.
- **Condition and column labels** must match the published codebook exactly. A validator script (R and Python) is provided in the tournament repository — teams run it before depositing; submissions failing validation at the lock date are scored only on the cells that parse.
- **One primary submission per team;** secondary submissions clearly labeled in the metadata file.
- **No revisions after lock.** Versioning before the lock is fine (hence `v<n>`); only the latest pre-lock deposit counts.

metadata.json

```json
{
  "team_id": "vienna",
  "team_name": "WU Behavioral Clones Lab",
  "contact": "name@institution.edu",
  "tier": 1,
  "entry": "primary",
  "approach_family": "per-respondent simulation, single model",
  "models": ["gpt-4o-mini-2024-07-18"],
  "registration_doi": "10.5281/zenodo.XXXXXXX",
  "disclosure_class": "A",
  "escrow_doi": null,
  "prediction_files": [
    { "file": "vienna_T1_primary_v1.csv",
      "sha256": "9f2c…e41a" }
  ],
  "coverage": { "interventions": 16, "outcomes": 13 },
  "blinding_attestation": true
}
```

(entry is "primary" or "secondary-1", "secondary-2", …; disclosure_class is "A" open, "B" escrowed, or "C" sealed; escrow_doi names the restricted-access record for Class B/C entries.)

Tier 1 — individual-level dataset

One row per synthetic respondent, matching the structure of the cleaned analysis data (cleaned_llm.rds schema). Each team supplies its own synthetic respondents: profile_id is a team-assigned identifier, unique within the submission, and assignment of respondents to conditions (between-subjects, covering all 17 conditions) is the team's responsibility, registered under C.4/C.6.

Column(s)	Type	Specification
profile_id	string	A unique respondent identifier you assign (unique within the submission).
condition	factor	Control or one of the 16 intervention labels, exactly as in the codebook.
gender, age_band, race, education, income, party	factor	The six moderators, levels per codebook (from your own profiles).
trust_multidimensional + 12 trust items	numeric	Primary composite 0–100; the 12 underlying items per codebook scale.
trust_post, distrust_post, funding_perceptions, policy_role_mean, inst_trust_mean	numeric	Secondary outcomes, original scales per codebook.
belief_post, concern_mean, policy_general, policy_specific_mean, behavior_mean	numeric	Tertiary outcomes, original scales per codebook.
donation_ams	numeric	Donation to AMS in $, per codebook.
newsletter_signup	0 / 1	Binary behavioral outcome.

Because a Tier 1 submission has the same shape as our own clone data, it flows through the original analysis pipeline unchanged and is eligible for **all** analyses. The model functions in R/functions/statistics.R are agnostic to the specific condition labels and to the data source, so the same code path applies to human data, our own clones, and any team's submission.

Tier 2 — cell-level file(s)

Two long-format files. `*_cells_main.csv`: one row per condition × outcome. `*_cells_moderator.csv`: one row per condition × moderator × moderator-level × outcome.

Column	Type	Specification
`condition`	factor	Per codebook.
`moderator`, `moderator_level`	factor	Moderator file only; per codebook levels.
`outcome`	string	Outcome variable name per codebook.
`mean`	numeric	Predicted cell mean on the original scale (proportion for `newsletter_signup`).
`sd`	numeric	Predicted within-cell SD (`NA` for the binary outcome).
`n_eff`	integer	Effective number of simulated respondents behind the cell.

Tier 3 — effect-level file

One row per intervention × outcome ($16 \times 13 = 208$ rows for full coverage). Estimates are ATEs versus control on the original outcome scale; the binary outcome is on the probability scale.

Column	Type	Specification
`condition`	factor	Intervention label per codebook.
`outcome`	string	Outcome variable name per codebook.
`ate`	numeric	Predicted ATE, original units.
`pi_lower`, `pi_upper`	numeric	95% prediction interval bounds — the team's epistemic uncertainty about the *human* ATE, not a within-simulation standard error. Used for inferential-agreement and equivalence scoring.

Analysis plan

Per-team analysis (reused)

Scoring follows the preregistered analysis plan without modification. The human sample is split in half by a preregistered random seed: every submission is scored against **Human 1**, and the agreement between Human 1 and Human 2 defines the **human–human ceiling**—the upper bound any predictive method could reach—reported in the same row format as every team. We do not restate the original plan here; see the original preregistration for the full specification of:

- **Section 1 — Average Treatment Effects:** ATE recovery across all outcomes (`run_main_treatment_model()` / `run_main_treatment_model_binary()`, compared with `compare_estimates()` and `run_calibration()`), and control-condition response distributions (`compare_distributions()`).
- **Section 2 — Subgroup Effects:** condition × moderator heterogeneity across the six moderators (`run_moderator_model()`), and within-subgroup distributions.
- **Section 3 — Demographic Baseline Calibration:** group-mean calibration (`compare_demographic_baselines()`) and demographic predictability / stereotyping (`compare_demographic_predictability()`).

Which of these analyses applies to a given submission follows from its tier (see *Analysis eligibility by tier*). One fixed **reference row** appears in every results table: the human–human ceiling (Human 1 vs. Human 2). The core team's own clone simulations (GPT-4o mini; Gemini 2.5 Flash Lite), registered with the original preregistration, enter the tournament as one team's submission under identical rules—they appear on the leaderboard like any other team and receive no special reference row.

Tier-specific estimand recovery

The comparison-metric ladder requires, per intervention × outcome, an estimate, a standard error, and an inferential category. How these are obtained depends on the tier:

- **Tier 1:** the submission runs through `run_main_treatment_model()` exactly like our own clone data; estimates, HC2 robust standard errors, and BH-adjusted p-values fall out of the fitted models.
- **Tier 2:** the ATE for each intervention × outcome is the submitted cell mean minus the submitted control mean. Because only cell summaries are available, the preregistered OLS cannot be refit on a Tier 2 submission; instead we use the standard two-sample (Welch) comparison, which is what the OLS treatment contrast reduces to in a between-subjects design. The standard error is computed from the submitted cell SDs and effective Ns, $\text{SE} = \sqrt{sd_t^2/n_t + sd_c^2/n_c}$, and the test statistic $z = \text{ATE}/\text{SE}$ is referred to the standard normal distribution for a two-sided p-value, $p = 2\,\Phi(-|z|)$. (With effective cell sizes in the hundreds, the normal approximation to the t distribution is essentially exact.) These p-values are BH-adjusted within outcome, mirroring the Tier 1 models. Subgroup ATEs and demographic baselines are computed analogously from the moderator cell file.
- **Tier 3:** the submitted `ate` is the estimate. Inferential agreement uses the submitted 95% prediction interval: an effect counts as significantly positive (negative) if the interval lies entirely above (below) zero—the direct analog of a two-sided test at $= .05$, so the

significance criterion matches the one applied to Tiers 1–2. For the TOST equivalence test, the interval implies a standard error, $\mathrm{SE} = (\mathrm{pi}_{upper} - \mathrm{pi}_{lower})/(2 \times z_{0.975})$.

Cross-team comparison (new)

This is the new analytical contribution of the amendment. We assemble a **leaderboard**: each submission is scored with the same metric ladder used throughout the original preregistration—in increasing order of strictness: directional agreement, Spearman , Pearson r, inferential agreement, RMSE, and TOST equivalence—so that approaches can be compared directly and against the human–human ceiling. Knowing on which rung an approach breaks is more informative than any single pass/fail verdict. On top of the ladder: the calibration regression (human ATEs on predicted ATEs) separates additive bias ( ) from proportionality ( ); distributional metrics (variance ratio, overlap coefficient, KS distance) test whether Tier 1 submissions reproduce response *shapes*, not just means; and the stereotyping diagnostics ask whether demographic variables predict synthetic responses with the same magnitude as in humans—or whether simulated respondents over-rely on demographic cues.

The headline leaderboard pools ATE pairs across all continuous outcomes (pooled RMSE is not reported, as it is not comparable across scales; per-outcome RMSE is reported in the per-outcome breakdowns); per-outcome, distributional, subgroup, and demographic-calibration columns follow for the tiers eligible for them, and every row carries its disclosure-class badge (open / escrowed / sealed) so readers see at a glance how much of each entry is auditable. Registered approach characteristics (open- vs. closed-weight, persona source, administration protocol, fine-tuned vs. not, …) are joined to the leaderboard for a descriptive, hypothesis-generating analysis of which design choices accompany better predictions—preregistered as exploratory, and restricted to publicly disclosed features (Class C entries are excluded from this analysis, not from the leaderboard).

> **i** Note
>
> **Exploratory by design.** With a modest number of teams, the relationship between approach characteristics and predictive performance is reported **descriptively** (rankings, group comparisons, visual patterns). We do not preregister inferential tests of feature effects; the cross-team feature analysis is hypothesis-generating, intended to surface which design choices appear to matter for future, larger benchmarks.

The remainder of this section demonstrates the leaderboard code on simulated multi-team placeholder data: five mock "teams", each a full LLM-style simulation drawn with a different seed and stacked with a `team` label. To exercise all three tier code paths, three mock teams enter at Tier 1 (individual-level), one is collapsed into a Tier 2 cell-level file, and one into a Tier 3 effect-level file. The code will be applied to real submissions without modification.

```r
outcomes_primary   <- "trust_multidimensional"
outcomes_secondary <- c("trust_post", "distrust_post",
                         "funding_perceptions", "policy_role_mean", "inst_trust_mean")
outcomes_tertiary  <- c("belief_post", "concern_mean", "policy_general",
                        "policy_specific_mean", "behavior_mean")
outcomes_continuous <- c(outcomes_primary, outcomes_secondary, outcomes_tertiary)

outcome_label_map <- c(
  trust_multidimensional = "Multidimensional trust (primary)",
  trust_post             = "Overall trust",
  distrust_post          = "Distrust",
  funding_perceptions    = "Funding perceptions",
  policy_role_mean       = "Policy role",
  inst_trust_mean        = "Institutional trust",
  belief_post            = "Climate belief",
  concern_mean           = "Climate concern",
  policy_general         = "General climate policy",
  policy_specific_mean   = "Specific climate policy",
  behavior_mean          = "Pro-climate behavior"
)
```

```r
# Extract ATE pairs (estimate, SE, adjusted p) - same helper as the original
# preregistration's rq1-all chunk.
prep_ate <- function(df) {
  df |>
    select(condition, estimate) |>
    mutate(
      se    = if ("std.error"        %in% names(df)) df$std.error        else NA_real_,
      p_adj = if ("p.value_adjusted" %in% names(df)) df$p.value_adjusted else NA_real_
    )
}

# Inferential category from estimate sign + adjusted p-value (Tiers 1-2)
classify_sig <- function(estimate, p_adj) {
  case_when(
    !is.na(p_adj) & p_adj < .05 & estimate > 0 ~ "pos",
    !is.na(p_adj) & p_adj < .05 & estimate < 0 ~ "neg",
    TRUE                                        ~ "ns"
  )
}

# Reference (Human 1) ATE models, fit once and reused for every comparison.
```

```r
human_models <- map(outcomes_continuous,
                    ~ run_main_treatment_model(human_data_1, outcome = .x)) |>
  set_names(outcomes_continuous)

human_side <- map_dfr(outcomes_continuous, function(out) {
  prep_ate(human_models[[out]]) |>
    rename(estimate_h = estimate, se_h = se, p_adj_h = p_adj) |>
    mutate(outcome = out)
})

control_lab <- levels(human_data$condition)[1]
z95 <- qnorm(0.975)

# Tier 1: pool ATE pairs from an individual-level dataset by refitting the
# preregistered treatment models on the submission.
build_pairs_t1 <- function(pred_data) {
  map_dfr(outcomes_continuous, function(out) {
    prep_ate(run_main_treatment_model(pred_data, outcome = out)) |>
      mutate(outcome = out)
  }) |>
    mutate(infer_l = classify_sig(estimate, p_adj)) |>
    rename(estimate_l = estimate, se_l = se) |>
    select(-p_adj) |>
    inner_join(human_side, by = c("condition", "outcome"))
}

# Tier 2: ATEs and SEs from submitted cell statistics (mean, sd, n_eff per
# condition x outcome); normal-approximation p-values, BH-adjusted within
# outcome to mirror the Tier 1 models.
build_pairs_t2 <- function(cells_main) {
  cells_main |>
    mutate(condition = as.character(condition)) |>
    group_by(outcome) |>
    group_modify(function(cells, key) {
      ctrl <- cells |> filter(condition == control_lab)
      cells |>
        filter(condition != control_lab) |>
        mutate(
          estimate_l = mean - ctrl$mean,
          se_l       = sqrt(sd^2 / n_eff + ctrl$sd^2 / ctrl$n_eff),
          p_adj      = p.adjust(2 * pnorm(-abs(estimate_l / se_l)), method = "BH"),
          infer_l    = classify_sig(estimate_l, p_adj)
```

```r
      ) |>
      select(condition, estimate_l, se_l, infer_l)
  }) |>
  ungroup() |>
  inner_join(human_side, by = c("condition", "outcome"))
}

# Tier 3: submitted ATEs taken as-is; inference from the submitted 95%
# prediction interval (significant iff the interval excludes zero, the direct
# analog of a two-sided test at alpha = .05); implied SE from the interval
# width for the TOST computation.
build_pairs_t3 <- function(effects) {
  effects |>
    mutate(
      condition  = as.character(condition),
      estimate_l = ate,
      se_l       = (pi_upper - pi_lower) / (2 * z95),
      infer_l    = case_when(
        pi_lower > 0 ~ "pos",
        pi_upper < 0 ~ "neg",
        TRUE         ~ "ns"
      )
    ) |>
    inner_join(human_side, by = c("condition", "outcome")) |>
    select(condition, outcome, estimate_l, se_l, infer_l,
           estimate_h, se_h, p_adj_h)
}

# Metric ladder over pooled pairs - mirrors compare_estimates() / the
# pooled_row logic in the original preregistration. Pooled RMSE omitted (not
# comparable across outcome scales).
pooled_metrics <- function(pairs) {
  pairs |>
    mutate(
      delta      = 0.5 * abs(estimate_h),
      diff       = estimate_h - estimate_l,
      se_diff    = sqrt(se_h^2 + se_l^2),
      p_tost     = pmax(pnorm((diff + delta) / se_diff, lower.tail = FALSE),
                        pnorm((diff - delta) / se_diff)),
      equivalent = p_tost < 0.05,
      same_sign  = sign(estimate_h) == sign(estimate_l),
      infer_h    = classify_sig(estimate_h, p_adj_h)
```

```
    ) |>
    summarise(
      directional_pct = mean(same_sign, na.rm = TRUE) * 100,
      spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                            use = "pairwise.complete.obs"),
      pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
      inferential_pct = mean(infer_h == infer_l, na.rm = TRUE) * 100,
      tost_pct        = mean(equivalent, na.rm = TRUE) * 100
    )
}

# One leaderboard row: pooled ladder + calibration slope and R^2 on the
# primary outcome. Works identically for all tiers since the calibration
# regression only needs (condition, estimate) pairs.
leaderboard_row <- function(pairs, submission, tier, disclosure) {
  calib <- run_calibration(
    human_models[[outcomes_primary]],
    pairs |> filter(outcome == outcomes_primary) |>
      select(condition, estimate = estimate_l)
  )
  pooled_metrics(pairs) |>
    mutate(submission = submission, tier = tier, disclosure = disclosure,
           beta = calib$beta, calib_r2 = calib$r_squared)
}
```

```
# Mock Tier 2 submission: collapse one placeholder team to the cell-level
# schema (*_cells_main.csv).
cells_t2 <- llm_data_teams |>
  filter(team == "team_4") |>
  select(condition, all_of(outcomes_continuous)) |>
  pivot_longer(-condition, names_to = "outcome") |>
  group_by(condition, outcome) |>
  summarise(
    mean  = mean(value, na.rm = TRUE),
    sd    = sd(value, na.rm = TRUE),
    n_eff = sum(!is.na(value)),
    .groups = "drop"
  )

# Mock Tier 3 submission: ATEs from one placeholder team with widened 95%
# prediction intervals standing in for a team's epistemic uncertainty.
effects_t3 <- map_dfr(outcomes_continuous, function(out) {
```

```r
  run_main_treatment_model(llm_data_teams |> filter(team == "team_5"),
                           outcome = out) |>
    transmute(
      condition,
      outcome  = out,
      ate      = estimate,
      pi_lower = estimate - z95 * 2 * std.error,
      pi_upper = estimate + z95 * 2 * std.error
    )
})
```

```r
# Three mock Tier 1 teams (mock disclosure classes shown to demonstrate the
# badge column), one Tier 2 team, one Tier 3 team.
team_rows <- bind_rows(
  leaderboard_row(build_pairs_t1(llm_data_teams |> filter(team == "team_1")),
                  "Team 1", tier = "1", disclosure = "A"),
  leaderboard_row(build_pairs_t1(llm_data_teams |> filter(team == "team_2")),
                  "Team 2", tier = "1", disclosure = "B"),
  leaderboard_row(build_pairs_t1(llm_data_teams |> filter(team == "team_3")),
                  "Team 3", tier = "1", disclosure = "C"),
  leaderboard_row(build_pairs_t2(cells_t2),
                  "Team 4", tier = "2", disclosure = "A"),
  leaderboard_row(build_pairs_t3(effects_t3),
                  "Team 5", tier = "3", disclosure = "A")
)

# Fixed reference row: the human-human ceiling (Human 1 vs. Human 2). The
# core team's own clones enter as a regular team submission like any other,
# with no special reference row.
ceiling_row <- leaderboard_row(build_pairs_t1(human_data_2),
                               "Human-human ceiling", tier = "-", disclosure = "-")

leaderboard <- bind_rows(ceiling_row, team_rows) |>
  arrange(desc(submission == "Human-human ceiling"), desc(pearson_r))
```

```r
leaderboard |>
  select(submission, tier, disclosure, directional_pct, spearman_rho, pearson_r,
         inferential_pct, tost_pct, beta, calib_r2) |>
  gt() |>
  fmt_number(columns = c(spearman_rho, pearson_r, beta, calib_r2), decimals = 2) |>
  fmt_number(columns = c(directional_pct, inferential_pct, tost_pct), decimals = 0) |>
  cols_label(
```

```
    submission      = "Submission",
    tier            = "Tier",
    disclosure      = "Class",
    directional_pct = "Directional %",
    spearman_rho    = "Spearman  ",
    pearson_r       = "Pearson r",
    inferential_pct = "Inferential %",
    tost_pct        = "TOST %",
    beta            = "Calib.  ",
    calib_r2        = "Calib. R²"
  ) |>
  tab_style(
    style     = cell_fill(color = "#eef3f7"),
    locations = cells_body(rows = submission == "Human-human ceiling")
  )
```

Table 7: Cross-team leaderboard (placeholder data). Pooled ATE-recovery metrics across all continuous outcomes, plus calibration slope and R^2 on the primary outcome, scored against the Human 1 reference half. The Human–human ceiling (Human 1 vs. Human 2) is the upper bound for any approach. Metrics run least-to-most strict, left to right; Class is the disclosure-class badge (A open / B escrowed / C sealed).

Submission	Tier	Class	Directional %	Spearman	Pearson r	Inferential %	TOST
Human–human ceiling	—	—	48	-0.10	-0.06	98	
Team 5	3	A	58	0.27	0.33	99	
Team 1	1	A	52	0.06	0.11	99	
Team 2	1	B	42	-0.15	-0.10	99	
Team 3	1	C	50	-0.07	-0.11	99	
Team 4	2	A	50	-0.13	-0.18	99	

```
ceiling_r <- leaderboard |> filter(submission == "Human-human ceiling") |> pull(pearson_r)

leaderboard |>
  filter(submission != "Human-human ceiling") |>
  mutate(submission = fct_reorder(submission, pearson_r)) |>
  ggplot(aes(x = pearson_r, y = submission)) +
  geom_vline(xintercept = ceiling_r, linetype = "dashed", colour = "grey40") +
  geom_segment(aes(x = 0, xend = pearson_r, yend = submission), colour = "grey80") +
  geom_point(size = 3, colour = met.brewer("Egypt")[2]) +
  annotate("text", x = ceiling_r, y = 0.6, label = "Human-human ceiling",
```

```
             hjust = 1.05, size = 3, colour = "grey40") +
  labs(x = "Pearson r (pooled ATEs vs. Human 1)", y = NULL) +
  theme_minimal()
```

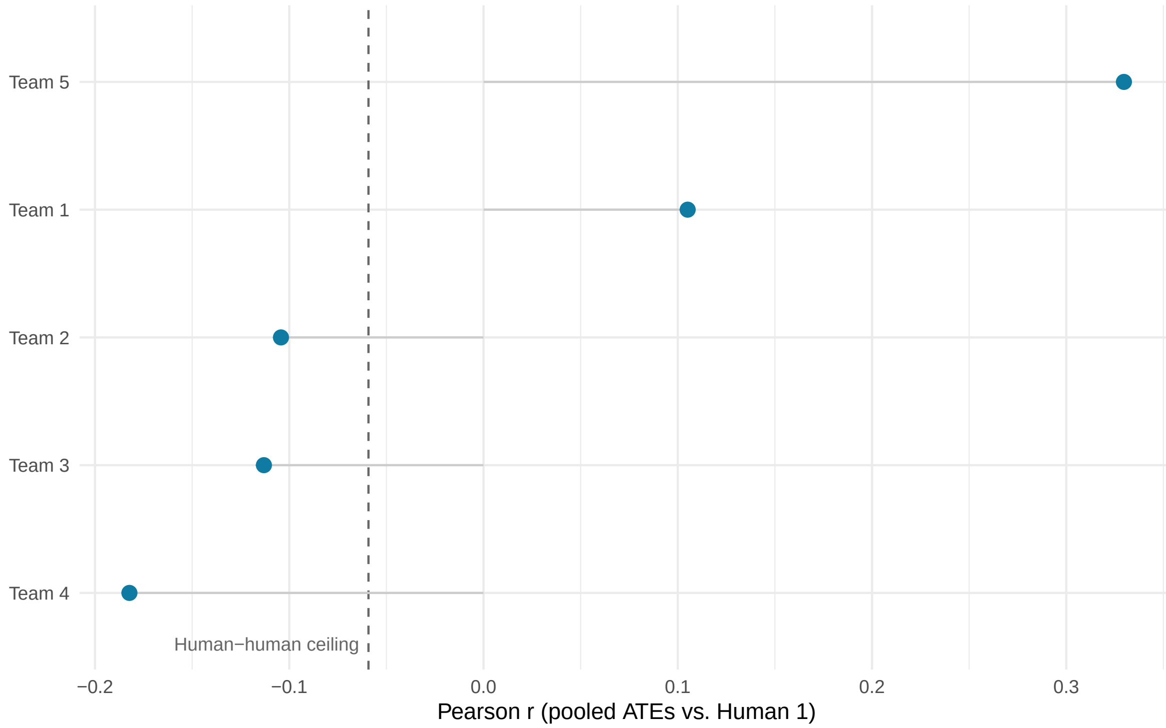

Figure 1: Approaches ranked by ATE-recovery (Pearson r of pooled ATEs against Human 1), with the human–human ceiling marked. On placeholder data all submissions are random noise, so no approach is expected to separate from the others.

When real submissions arrive, this leaderboard is recomputed by replacing the placeholder datasets with the submitted prediction files routed through their tier's builder (`build_pairs_t1()` / `build_pairs_t2()` / `build_pairs_t3()`), and by adding the distribution, subgroup, and demographic-calibration metrics from Sections 1b, 2, and 3 as additional leaderboard columns for the tiers eligible for them. The approach-characteristics table coded from the registration forms is then joined to the leaderboard for the exploratory analysis of which design choices accompany better predictions (publicly disclosed features only; Class C entries excluded from this analysis, not from the leaderboard).

Cummins, Jamie. 2025. "The Threat of Analytic Flexibility in Using Large Language Models to Simulate Human Data." arXiv. https://doi.org/10.48550/ARXIV.2509.13397.

Feuerriegel, Stefan, Christopher Barrie, M. J. Crockett, Laura K. Globig, Killian L. McLoughlin, Dan-Mircea Mirea, Arthur Spirling, et al. 2026. "A Reporting Checklist for Large Language Models in Behavioural Science." *Nature Human Behaviour*, June, 1–5. https://doi.org/10.1038/s41562-026-02492-7.